Memory & Cognition
https://doi.org/10.3758/s13421-025-01803-z

The five Garners: The psychological contributions of Professor Wendell R. Garner

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Received: 28 January 2025 / Accepted: 21 September 2025
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Abstract
In the domain of perception and cognition, few scientists have contributed more broadly and profoundly than W. R. Garner. Today, Garner is best known for the eponymous *Garner interference*, but several equally epoch-making developments are less widely recognized or associated with his name. In this article, we provide an in-depth examination of five areas developed by Garner's pioneering ideas, all precursors of contemporary research. They include hearing psychophysics, methodology (with implications for resolving the replication/credibility crisis), information theory, perceptual independence, and attention. All have been instrumental in shaping and molding current perceptual and cognitive science. This article can also serve the student of cognition and perception as a reference for the origins of contemporary work and findings in psychological science.

Keywords Hearing · Information theory · Gestalt · Independence · Selective attention

Introduction

Wendell R. Garner (1921–2008) made profound contributions to multiple domains of behavioral science in the course of his storied career. His work revolutionized the way we think about perception and cognition and continues to wield pervasive influence on current research and theory. An unfortunate by-product of his wide range has been that students familiar with Garner's pioneering work in one domain often are unaware of further breakthroughs in other domains. This range thwarted recognition of Garner's extraordinary stature. The goal of this article is to provide an outline of Professor Garner's many contributions within a single framework. The review also follows the evolution of Garner's thinking that guided his transformative science and movement to ever-new frontiers.

Let us illustrate his range with Garner's response to a remark made by the first author during one of their last meetings. Algom noted, "You know Tex (the nickname he never liked), there are three Garners," to which he retorted, "You are wrong Danny, there are five Garners!" He was right of course (see also Algom, 2016, on that). For another illustration, the Garner listed in the Yale catalogue of 1979 taught classes on attention, structure, and Gestalt. This contrasted sharply with the Garner listed in works of hard-core psychophysics much earlier (e.g., Garner, 1947, 1948, 1954a, 1954b; Garner & Miller, 1944, 1947; see Algom & Babkoff, 1984). Were these the same Garner? They were indeed! The illustration highlights the tectonic shift in Graner's research from advanced auditory psychophysics to the terra incognita at the time of simple visual shape and dot patterns. So much for range.

Today, Garner is probably best known for what is called "Garner interference" (Pomerantz, 1983) and the related "separability-integrality distinction" (Algom & Fitousi, 2016), but these are just two of a series of profound contributions. As we sketch below, Garner's contributions span the gamut from psychophysics to methodology to information theory to perceptual independence to Gestalt and to selective attention. This is the structure of our review. First, we detail the evolution of Garner's thinking embedded within the main landmarks of his life. We then devote separate sections to discussing, in turn, psychophysics and hearing, methodology, information theory, figural goodness and Gestalt, perceptual independence, and attention.

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Published online: 09 October 2025

Synopsis of a remarkable opus: The life and times of Wendell Garner

Wendell Richard Garner was born on January 21,1921, in Buffalo, New York and died on August 14, 2008, in Redding, Connecticut. Starting at Harvard in 1943, he devoted some 50 years to studying psychological science. During the war, he worked at the Radio Research Laboratory in Tennessee and subsequently in the Psychoacoustics and Systems Research Laboratories in Massachusetts and in Rhode Island. One of his reports, tentatively called "Factors Involved in the Randomization of the Radar PRF," subsequently served to support his Ph.D. under the direction of S. S. Stevens, awarded in 1945. The report was classified top secret, the sealed document inaccessible to everyone (including the examiners). Shortly afterward, Garner began his 22-year association with the Department of Psychology at The Johns Hopkins University. During those years, he served as the Director of the Psychological Laboratories and the Institute for Cooperative Research, as well as the Chairperson (1954–1964) of the Department of Psychology (Fig. 1).

As we hinted, Garner's first domain of study was psychological acoustics or the psychophysics of hearing. This was BCE – before the computer era – so that acoustical research entailed added technical and mathematical skills on the part of the experimenter. Tackling the challenges, Garner developed the *lambda scale* of loudness based on the converging results from the methods of fractionation and sense distance. Another fundamental contribution during the Johns Hopkins years was the determination in hearing of temporal integration, a phenomenon previously documented only in vision. The quality of Garner's work secured a steady stream of publications in the most prestigious and selective outlet in the field, *The Journal of the Acoustical Society of America* (shared by physicists and psychophysicists). More work was published in the *Journal of Experimental Psychology*.

We mention those top journals to highlight the contrast with the marginalized outlets where Garner was forced to publish for some time after his bold move away from psychoacoustics. This sharp change of course was partly motivated by Garner's disillusionment with Stevens' (1956, 1975) method of Magnitude Estimation that engulfed the field. To Garner's mind, presenting people with a sound and recording the number they offer for its loudness does not constitute measurement. Garner's opposition to the so-called "direct" methods of measurement (those without a conceptual infrastructure to support theory) remained steadfast throughout.

We pointed out that the *lambda scale* of loudness is based on the converging results of separate methods of measurement. This is notable. The concept of converging operations was introduced in the breakthrough article, "Operationalism and the concept of perception" (Garner et al., 1956), and the idea remains a cornerstone of psychological research to date. Multi-lab multi-method operations are the key to resolving the current replication/credibility crisis. The designation of the outcome by a single-lab single-method operation is just another name for an experimental result, lacking any theoretical content.

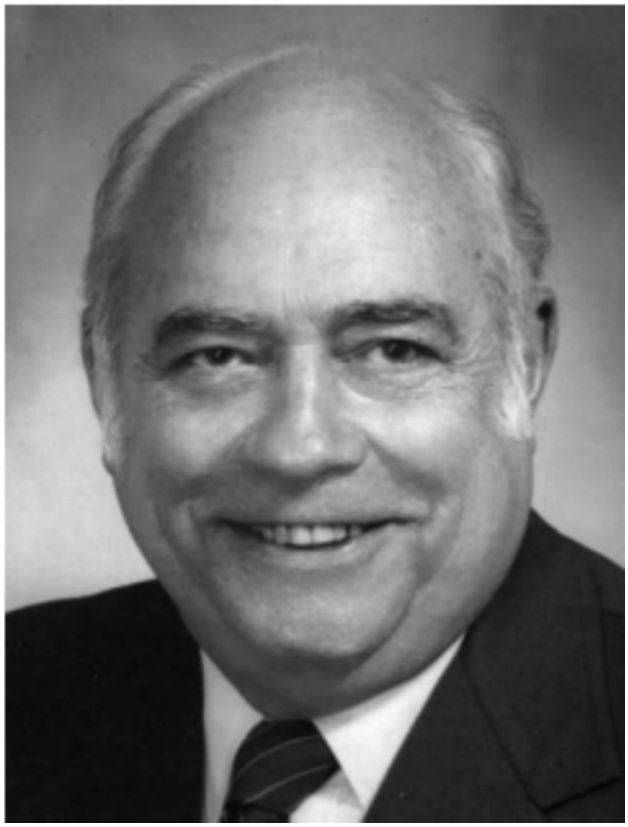


Fig. 1 Wendell R. Garner (during his Hopkins years)

Simultaneously with his departure from psychophysics, Garner became interested in the still inchoate domain of psychological information theory, the applications to psychology of Shannon's (1948) formulation of "information theory." A particularly appealing aspect of the pertinent methods was their nonmetric nature. An early fruit of this interest (already in Garner's acoustical era) was the development of *Absolute Identification* – a method free of "numerical" estimates by the observer. However, the true culmination of that interest was the epoch-making, "Uncertainty and structure as psychological concepts" (Garner, 1962), a book that changed the field for good. The hallmark of Garner's contribution was the foundational nature of the work. Garner's ideas are the bedrock

foundation of all modern work on information theory in psychology (Fitousi, 2013, 2025a, 2025b).

One more item from the Hopkins years is worth mentioning. Early in his tenure Garner published a textbook on methodology that pioneered the use of *system analysis*, "Applied experimental psychology: Human factors in engineering design" (Chapanis et al., 1949). The book was required reading in programs of psychology in the ensuing decades. System and structure were to remain the hallmark of Garner's research career.

When Garner left for Yale in 1967, he was already a full-fledged *cognitive* psychologist. His research focused on figural goodness, the linchpin of the early Gestalt efforts to revive the ideas on aesthetics and beauty cultivated in ancient Greece (Mach, 1886/1959; Weyl, 1952). Amidst the numerous Gestalt "laws," Garner developed a formal theory that allows us to define figural goodness in an objective and quantifiable manner (Garner, 1970, 1974). Subsequently it was shown that Garner's formalism is isomorphic with mathematical group theory (Palmer, 1991; see also, Algom & Fitousi, 2025), but Garner's development is distinguished by its close affinity with information theory. Apart from his influence on the as-yet underdeveloped field of perception of the arts, Garner's development aligns well with modern research in the domain of randomness and complexity. Recently, Algom and Fitousi (Algom & Fitousi, 2025; Fitousi & Algom, 2024) found significant correlations between the Garnerian values of figural goodness and advanced measures of algebraic complexity.

Concurrently with the studies on figural goodness, the notion of perceptual independence attracted interest. This research, too, grew out of Garner's earlier pursuits in information theory. Are different features of a stimulus or of two different stimuli processed independently in the perception? There were few methods in the toolkit of the cognitive psychologist to even start tackling this fundamental question. On that background, Garner, in a pioneering effort, conceived the first experimental designs and analytical routines tailored to decide the issue in any particular instance (Garner, 1962; Garner, 1970).

Finally, in the framework of the quest for capturing independence, Garner conceived the distinction of *separable vs. integral* dimensions, a pillar of current cognitive science. The former are processed independently of one another; the latter are prone to intrusions from one dimension into the other. An evolved experimental design, "Garner's speeded classification paradigm" (Garner, 1974, 1976), was developed to decide the status of the pair of dimensions under test. One particular effect within the paradigm, "Garner interference" (Pomerantz, 1983), has attracted a great deal of attention, rivaling the well-known Stroop effect (Algom & Fitousi, 2016). By employing *speeded* classification with the attendant time-recording apparatus, Garner's research seems to have come full circle in terms of gadgetry.

The main point that emerges from our synopsis is the rich yields of Garner's manifold research. Amid the great richness and fecundity, our synopsis is still selective. Left out, for example, is Garner's pioneering research on baseball (several decades ahead of Tversky's similar venture into data analysis in sports, e.g., Tversky & Gilovich, 1989). Garner used to note, tongue-in-cheek, that the publication on baseball was his most-read piece (Cook & Garner, 1964). Similarly left out are studies on temporal patterns, learning of correlational structures, or free classification, among other lesser subjects. Focusing on the five main contributions, is there a dominant theme running through all the various attainments? Along with Garner (1975, 1991), we believe there is. The concept structure unifies the various contributions. All subsequent research on information and structure in psychology bears the imprint of Garner.

During his career, Garner was awarded many honors including election to the *Society of Experimental Psychologists* (SEP) and receipt of their *Howard Crosby Warren Medal*. He was named a member of the *National Academy of Sciences*. From the *American Psychological Association* he received the Distinguished Scientific Contribution award and from the American Psychological Foundation their *Gold Medal Award for Life Achievement in Psychology*. Perhaps the most satisfying of all was the Festschrift organized in his honor by former students Lockhead and Pomerantz. The conference and celebration drew to Yale the best and brightest scientists in the field. The American Psychological Association published the contributions, and the book was aptly titled *The perception of structure* (Lockhead & Pomerantz, 1991).

Garner shared his remarkable career with his wife of 62 years, Barbara Garner. Barbara was always supportive during their long and happy life together. This scientific success story along with many years of marital bliss is all the more striking when we consider Garner's laborious youth, enduring incredible hardships. He was a child of the Great Depression, experiencing extreme poverty, abandonment by parents, and homelessness. During that difficult time, he contracted polio, which affected his speech throughout the years.[1]

We now are ready to start our polymorphous journey around the evolution of Garner's research – from cutting edge technology to paper and pencil, and back (up to a point).

[1] When the subject of hardship was brought up during their last meeting, Garner whispered in Algom's ear: "Danny, I am happy with my life!".

Garner #1: Hearing, psychophysics, and scaling

During the first years of his scientific activity, Garner's interest focused on the psychophysics of hearing. This acoustic epoch, starting already at Harvard a few years before getting his Ph.D. (in 1945), lasted over a decade. Of several significant contributions, we discuss two here.

Temporal integration of acoustical energy

To facilitate understanding, let us consider temporal integration in vision first. Using a traditional (i.e., non-digital) camera containing a light-sensitive film, the reaction of the film (its absorbance expressed in the optical density of the developed film), hence the quality of the resulting picture, depends on total light energy (E). This energy is the product of exposure time (t) and intensity (I), or E = I x t. Users of the camera control duration via shutter speed and they control intensity via the aperture (hole) that limits the light entering the device. The multiplicative time-intensity relation means that one can reduce intensity while increasing duration to achieve the same result. In physics and photography, this reciprocal relation has been known as the Bunson-Roscoe law (Algom & Babkoff, 1984). Because the human eye contains visual pigments in the retina, the reciprocal relation of time and intensity applies to human light detection too. The multiplicative relation with respect to vision is known as Bloch's law (Algom & Babkoff, 1984). The Bunson-Roscoe law in physics and Bloch's law in human vision mean that light intensity is integrated over time up to some critical duration.

Does temporal integration exist in audition? Can extending the duration of a sound compensate for a drop in its intensity to preserve detection? Again, the reciprocal time-intensity relation or temporal summation had only been documented in vision. With virtually nothing known at the time in audition, it was Garner who introduced the phenomenon of auditory temporal integration into hearing science and provided its first methods of measurement and theoretical interpretation. In a series of studies (Garner, 1947, 1948, 1949), some done jointly with his classmate and roommate in Cambridge, George Miller (Garner & Miller, 1944, 1947), Garner uncovered the basic parameters of temporal summation in hearing. He established that, for pure tones (sinusoids of a given frequency), temporal integration of acoustic energy at threshold is linear. One can reduce sound intensity tenfold (attenuating sound pressure to the ear by 10 dB) yet retain the ability to detect the tone by increasing its duration tenfold. Garner further found that this perfect time-intensity reciprocity held up to a critical duration of 250 ms. Beyond the critical duration, time still affected detection partially up to 1–2 s. Beyond that, absolute threshold and loudness are determined only by sound intensity irrespective of duration (see also, Algom & Babkoff, 1978, 1984; Algom et al., 1980; Babkoff & Algom, 1976; Marks, 1974a, 1974b). The basic features of auditory temporal integration uncovered by Garner are illustrated in Fig. 2.

The general equation that best describes the data is $I \times t^a = C$, where C stands for threshold detection of the sound. The exponent a is the slope of the straight line relating log I (dB) to log t. When a = 1, there is a perfect trading relation between time and intensity. As we just recounted, this holds within a time window called critical duration, typically found at 250 ms in hearing (critical duration is appreciably smaller in vision, more like 100 ms). Beyond that window,

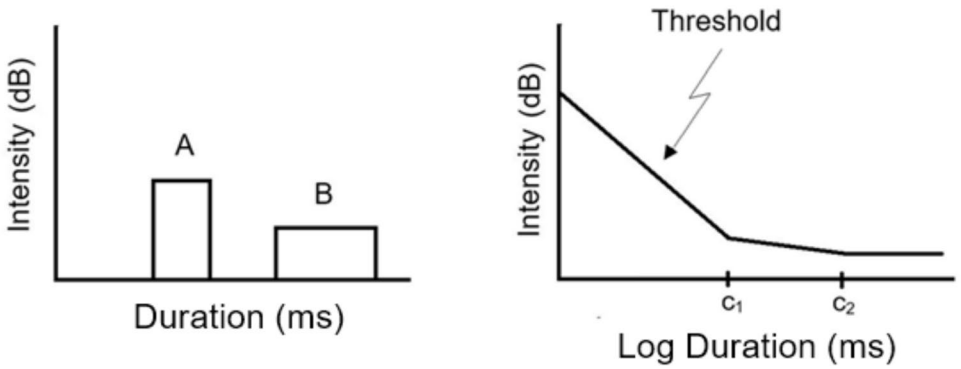


Fig. 2 Depictions of auditory temporal summation systematically studied by Garner. Left panel: The equal areas of the rectangles A and B represent the respective sound energies. They are the same regardless of whether the intensity of the sound is high (A) or low (B) and duration is short (A) or long (B). Right panel: The time-intensity tradeoff for sound detection. Up to a critical duration (c_1), a tenfold change in intensity can be offset by a corresponding tenfold change in duration to maintain the same threshold of hearing. Further increases in duration up to a second value (c_2) still affect threshold but cannot fully compensate for attenuation of intensity. Beyond c_2, only intensity determines threshold

$0 < a < 1$, meaning that time still affects detection, but that it cannot fully compensate for a reduction in intensity. Such partial integration can last into 1–2 s. Beyond that limit, threshold and loudness are determined solely by intensity. In addition to establishing these foundations, Garner probed the influence on summation of several auditory variables including frequency, noise, or masking. Reviewing those, however, would take us too far afield.

Notably, these studies were performed in the best tradition of classical psychophysics. For example, in the first Garner and Miller study (Garner & Miller, 1944), the participants were the authors themselves with their data listed separately and depicted individually in the graphical presentation. The experiment lasted 2 months (!), the first of which was devoted to training. These practices harken back to Wundt's if not to Fecher's laboratory, where, it is said, a participant would respond in 24,000 trials per experiment (Murray, 2021; see also, Miller, 1998). They also reflect the widely held assumption within psychophysics that the auditory apparatus (like any other sensory apparatus) is cognitively impenetrable so that the authors can safely serve in the role of observers, too (Murray, 2021).

Scale for loudness

To appreciate the value of Garner's work on loudness, the subjective sensation of sound intensity, a little background is necessary. How does loudness depend on the sound's physical intensity? The question was already discussed by eminent Greek scholars of antiquity, but its modern quantitative study is coextensive with the advent of the discipline of psychophysics by Gustav Theodor Fechner in 1860. It had long been known that subjective intensity is *not* a linear function of physical intensity. In his *Elemente der Psychphysik*, Fechner cites the virtuoso violinist, Wasilewski, who reported from a Rhine musical festival that the chorus of 400 singers did not sound louder than the chorus of 200 singers (Fechner, 1860, p. 182).

This observation is consistent with Fechner's epochmaking psychophysical law: The magnitude of sensation (ψ) is a logarithmic function of the magnitude of the physical stimulus (S), such that $\psi = m \log (S/S_0)$, where S_0 is the absolute threshold and m is a constant of measurement. Of equal importance (though probably less often discussed) are Fechner's methods of measurement establishing the law. They included procedures of bisection in which the observer sets a stimulus to lie halfway between two stimuli. In equisection, the observer adjusts several stimuli to mark off equal sense intervals. The observer can also rate a series of stimuli by consecutive numbers so that each number marks off a constant increment in sensation. Collectively, the methods are usually called category scaling (Marks & Algom, 1998).

What is particularly remarkable about the Fechnerian methods is what they do *not* entail: direct numerical judgments of magnitude. The observers are never asked to provide a number to stand for their sensation of the stimulus. Fecher's "indirect" methods are valuable precisely due to this feature because they naturally contain a strong metric structure to support validity. Now fast forward a century to S. S. Stevens (e.g., Stevens, 1975), who advocated, contra Fechner, "direct" methods where the observer provides a number (R) to stand for the magnitude of sensation (S). Stevens found that R was a power function of stimulus magnitude governed by an exponent (b), such that $R = kS^b$, where b is the exponent and k is a constant of measurement (see Algom, 2021). For auditory intensity, b was determined to equal 0.6, thereby defining Stevens's *sone scale* of loudness.

Enter Garner. Garner was not prepared to accept Stevens's "direct" method of magnitude estimation and the attendant sone scale as valid measurement. The number proffered by an observer in response to the stimulus might not be a genuine number obeying the axioms of arithmetic. That "number" may mean no more than saying "big" or "very big," in which case averaging and other arithmetic operations performed on these "numbers" are not justified. Moreover, the relation between those numbers R and the actual *sensations* ψ is unknown. Does R faithfully reflect ψ? We do not have evidence on this crucial question of validity. However, Stevens's method of magnitude estimation, the power law of sensation, and the *sone scale* of loudness are only valid if R *is* proportional to ψ, i.e., $R \propto \psi$. Investigators were slow to recognize this stipulation (Marks, 1974a) and, again, it has not been adequately addressed or corroborated to date (Algom, 2021). The upshot is, to Garner's mind, proffering a "number" in response to a single stimulus is not measurement.

In contrast to Stevens, Garner (1954a) based his research on the methods of bisection and equisection and derived a power scale for loudness with an exponent that is markedly smaller than that governing Stevens's *sone scale*. In Garner's *lambda scale*, b equals 0.26, so that loudness (L) in lambda units is $L = kp^{0.26}$, where p is the sound pressure fed to the ear. Now whether loudness relates to sound pressure via $L = kp^{0.6}$ (sone) or via $L = kp^{0.26}$ (lambda) makes a difference not merely for the psychophysical function for loudness but also for substantive processing of sounds in hearing. Consider the phenomenon of binaural summation, the experience that a tone heard by both ears sounds louder than the same tone heard by a single ear. In terms of sones, binaural summation is perfect, namely, binaural loudness (L_2) equals the simple algebraic sum of the individual loudness at each ear (L_1), so that $L_2 = 2L_1$ (e.g., Algom & Marks, 1984, 1990; Marks, 1978). By contrast, in terms of lambda, a sound presented to the two ears is only 1.4 times as loud, or $L_2 = \sqrt{2}L_1$ (Garner, 1954a, 1954b, 1958; see also, Marks,

1974b). A parallel difference is present when we examine the total loudness experienced when tones of different frequencies are played simultaneously (think of an orchestra). Counted in sone units, total loudness (L_T) is the linear sum of the individual loudnesses so that $L_T = \sum L$. When loudness is counted in units of lambda, by contrast, total loudness is the square root of the sum of squares of the loudnesses of the individual tones, so that $L_T = (\sum L^2)^{1/2}$ (e.g., Garner, 1959; Marks, 1974b).

Is the sone scale or the lambda scale the true scale for loudness? The most advanced psychophysical tools available to us still lack resolution power and precision to settle the issue (Marks, 1974a, suggested a possible reconciliation). What can be affirmed with confidence, though, is that it was Stevens's sone scale that carried the day.

We conclude by noting the present change of perspective on the issue. Garner's reservations were not lost. Their impact is fully felt in current psychophysics, so that Garner's reservations regarding "direct" methods are widely shared now (Algom, 2021). Garner's classic studies on loudness (e.g., Garner, 1954a, 1954b, 1958, 1959) are now six decades old, but they are stunningly modern in their insistence on the presence of a metric structure to sustain the evaluation of loudness. Garner's insistence on the need for an underlying metric structure is sine qua non for modern sensory measurement across psychological science.

In 1958, Garner completely closed down his auditory laboratory. That took courage because Garner had accrued a strong reputation in audition. Nevertheless, he was determined to change course even at the cost of throwing his reputation away. Many a scientist who tries to change their subject matter fails because they keep accepting invitations for speeches and have a safe path to publishing due to their reputation in the field that they wish to leave. Garner's own first test came with an invitation to contribute a chapter on audition for a major handbook. He thanked the editors, but stated that he no longer worked in audition.

Before reviewing Garner's new subject matter in vision, we must discuss a theme that attracted his attention along with hearing and that remained a mainstay of research throughout his career.

Garner #2: Converging operations

The maxim of positivism that the meaning of all scientific terms derives from "positive" data of experience, from observation, is widely accepted. The physicist Percy Bridgman crystallized this teaching of positivism into a manual of experimental *operations* (Bridgeman, 1927). According to Bridgman, all variables in a scientific study must be defined through measurable experimental observations or through the manipulations that produced the variable within the experiment. Garner, along with virtually all contemporary researchers, accepted these dictums. At the same time, Garner emphasized that there is a limit to what operational definitions and manipulations can do: They are indispensable for establishing the validity of a variable and its outcome in the experiment, but they cannot establish that outcome as a *theoretical concept*. To create a theoretical concept, converging operations are needed.

The idea of converging operations was introduced in the breakthrough article, "Operationalism and the concept of perception" (Garner et al., 1956), and it remains a cornerstone of psychological research to date. The core idea is best described by Garner himself (Garner, 1974, p.187): "a concept that is synonymous with a single operation is nothing more than a restatement of an experimental result, but a concept that arises as a consequence of converging operations has a reality that is independent of any single experimental observation." Converging operations are especially indispensable for defining theoretical notions that are unobservable experimentally. Consider the notion of "perceptual independence." This foundational notion is unobservable by any single experimental manipulation. Only by drawing on a variety of operations that converge on the same consequence can one explicate this *theoretical* concept. A powerful illustration of this tactic – employing experiments, conceptual analysis, and modelling – is the delineation of "independence" in Ashby and Townsend's General Recognition Theory (GRT; Ashby & Townsend, 1986; Kadlec & Townsend, 2014), as well as in Townsend's System Factorial Technology (SFT; Townsend & Nozawa, 1995; Little et al., 2017; see also Algom, 2016; Algom et al., 2015).

For another illustration, Garner, in the 1956 article, considered the notion of intelligence. The well-known definition that "intelligence… is what the intelligence tests test" (Peak & Boring, 1926) is a favorite quip in classes of psychology. Following the dictum of converging operations, this definition does not qualify as a construct of intelligence – "intelligence" in this example is simply another name for a numerical experimental result. For intelligence to evolve into a real theory, different experimental operations and observations are needed such that all converge on a common result.

Another ready example comes from the aforementioned method of magnitude estimation. In that method, the participant responds with a number that matches the magnitude of sensation upon presentation of a stimulus of a certain magnitude. Stevens considered those numbers as concepts that faithfully reflect sensations – without evidence. To accept that magnitude estimates are a genuine concept with theoretical content, an infrastructure of converging operations is needed. Absent of such operations, the method amounts to a single observation devoid of theoretical value.

For another example, consider Garner's pivotal distinction between integral and separable dimensions in perception

and attention (Garner, 1974; see also, Algom & Fitousi, 2016). Separability is a theoretical concept of great importance precisely due to the use of completely *different* methods that converge in supporting it. We develop this topic in the section on selective attention, but here we mention three independent methods that provide cross-validation support by converging on the same outcome. The main operation used for testing separability is Garner's speeded classification paradigm, which we mentioned earlier. Now if that was the only method used, separability would merely have amounted to another name for an experimental observation.

A second, completely distinct method, is based on the outcome of multidimensional scaling of the set of stimuli (Shepard, 1957). In particular, one examines special cases of the Minkowski distance that best fit the outcome. If the best-fitting metric is Euclidean, the dimensions are integral. If the best-fitting metric is city-block, the dimensions are separable. A third method, unconnected to the others, employs information theory. If identification improves by adding another dimension to the presented stimuli, then the original and the added dimensions are separable; if performance remains the same, the dimensions are integral. Typically, the three operations converge on the same outcome (Baird & Noma, 1978).

Let us issue an important caveat: The effect and notion of separability and the separability-integrality distinction naturally entail sundry nuances and contextual constraints. In that respect they emulate the Stroop effect and its many interpretations (MacLeod, 1991; Melara & Algom, 2003; Sabri et al., 2001; see also, Eidels, 2012). In a recent review, Algom and Fitousi (2016) discussed the various contextual constraints on the separability-integrality distinction in considerable detail, concluding that the notion of separability is a solid one underlying much contemporary research.

What operations do not qualify as converging operations? That class is called *parallel operations*. An example would be performing the experiment again but only with a sample of the original stimuli. Clearly, independent convergence is impossible in this case to sustain a theoretical concept. Another example is *repeat operation.* This consists of performing the same experiment at another time. This procedure – doing the same thing twice – lacks converging operations, too. If the original experiment manipulated a single variable, repeating it does not help in forming a theory. A third example is *transform operation.* In this method, a variable is added that is merely a transformation of the original variable. An example from numerical cognition would consist of presenting a new set of numbers for stimuli – each one is the square of the original. Or, in a verbal experiment with vulgar words, the new set would consist of synonyms of the original ones. In all such cases, the two variables and the attendant manipulations do not form truly independent operations.

The replication crisis and converging operations

The principle of converging operations has profound ramifications for the replication crisis experienced in psychology. The crisis came to the fore in the early 2010 s when it became clear that large portions of the results of published studies in psychology could not be replicated (e.g., Eronen & Bringmann, 2021; Lakens, 2023; Vazire, 2018). This situation has been attributed to questionable research practices, themselves the result of the unswerving pressure to publish. The dubious practices included: favoring data supporting the hypothesis (confirmation bias), small sample sizes (lacking statistical power), p-hacking, selective reporting ("cherry-picking"), or Harking (post hoc generation of hypotheses), among others (e.g., Ioannidis, 2005; Lakens, 2023; Lilienfeld, 2017). To address the problem, a massive shift in publication procedures has been taking place, the chief countermeasure being pre-registration of studies (Lindsay, 2020). The goal of this measure is to eliminate the most blatant biases while rendering all publishable data transparent.

For all their merits, the remedies, with their excessive emphasis on replicability, carry drawbacks, and hence are not likely to solve the current crisis (Vallesi, 2025). It is at this juncture that Garner's guidance on converging operations becomes crucial: Exclusive insistence on replicability of (one) method cannot be the ballast needed for cognitive science. This approach kills theory, which needs multiple converging evidence. Consider Vallesi's (2025, p. 3) conclusion: "An excessive focus on trying to empirically reproduce or replicate published effects or phenomena…, if an end in itself, can divert attention away from developing robust theories." Vallesi (2025) further concludes that the current means imposed on the publication process are bound to create a *theory crisis* in psychology – in lieu of the replicability crisis.

Again, recent criticisms of registration and pre-registration (e.g., Kaufman & Glaveanu, 2018; Ness, 2015) echo Garner. To establish a theory, one must perform a set of independent experimental procedures (converging on the same outcome). Replication by itself is not sufficient. At best, it produces a reliable variable or result – devoid of theoretical content. Replicating an experiment does not guarantee advance in psychological theory. If the original experiment lacked a solid theoretical construct, replicating it does not amount to much.

At this point we are ready to consider Garner's research and contributions in his newly chosen field of vision and information theory.

Garner #3: Information theory and multivariate analysis

Garner was among the first psychologists to realize the potential of Shannon's (1948) model of information theory for a new understanding of old problems. Among several pioneers (e.g., Attneave, 1959; Miller, 1953, 1956), Garner's approach still stands out for its development of a whole *system* and *terminology* (rather than single unconnected developments). The ideas that emerged from Garner's effort form the basis of the entire field of modern psychological information theory (Fitousi, 2013, 2025a, 2025b). We briefly discuss three related contributions.

The nature of information: Multivariate information analysis

Garner (1962) created, virtually ex nihilo, an entire system of psychological information augmented by a new vocabulary. We attempt to keep technical details at a minimum yet still provide the essence of this system.

Recall the fundamental formula of information (Shannon, 1948),

$$I(x) = \log\ N, \tag{1}$$

where I is information in *bits* (i.e., the basis of the log is 2) and N is the set of equiprobable alternative stimuli in which stimulus x resides. When all stimuli are not equally likely to occur and when we wish to know the long-term *average* information, we can write the formula in terms of U, or uncertainty (the reverse of information I),

$$U(x) = -\sum p(x)\log p(x). \tag{2}$$

We cannot overemphasize the fact that this (or the former) formula applies to *all* collections of events regardless of how the events are defined or manipulated by the experimenter. Given the nonmetric nature of information, there is no difference between a distribution of uncertainties in terms of a single variable and a distribution of the same uncertainties defined in terms of two or more variables. The measure of information or uncertainty is indifferent to the dimensional structure of the experiment. The difference is simply the way that one classifies the same data. However, there is still gain to be reaped from a multi-class partition – the emergence of a new set of useful terms, *joint uncertainty*, *maximum joint uncertainty*, or *contingent uncertainty* (among others). The meaning of these terms is best explained by the following example.

Suppose that we have a matrix of probabilities of occurrence defined by the joint values of variable x (consider it as the stimulus) and variable y (the response) as given in Table 1.

The uncertainty of each cell p(x,y) as well as their average uncertainty can be computed in exactly the same way as any (one-dimensional) series of uncertainties. Therefore, the average uncertainty of the matrix is:

$$U(x,y) = -\sum p(x,y)\log p(x,y). \tag{3}$$

Table 1 Matrix of probabilities of the joint occurrence of values of variables x and y

		Y			
		1	2	3	
X	1	0.24	0.06	0.06	**0.35**
	2	0.06	0.12	0.06	**0.24**
	3	0.06	0.06	0.29	**0.41**
		0.35	**0.24**	**0.41**	

This term, U(x,y) is called *joint uncertainty*. In our example, this value is 2.81 bits (which the reader is invited to verify). The fact that the probabilities of occurrence are not equal in the matrix indicates that there is a correlation between x and y. However, we can transform the data such that the two variables become orthogonal. All that we need to do is to change the probability in each cell to equal the cross-product of the appropriate marginal probabilities of the rows and the columns (the way that one calculates chi-square or the contingency coefficient). Doing so, the former matrix becomes that shown in Table 2.

We can calculate the average uncertainty in this second matrix in exactly the same way using the new probabilities, P(x,y). This calculation defines a new term, *maximum joint uncertainty*, or U_{max}:

$$U_{max}(x,y) = -\sum P(x,y)\log P(x,y). \quad (4)$$

The adjective *maximum* is added because the equation indicates the maximum uncertainty that can be obtained – limited by the given row and column marginal probabilities. It is the average uncertainty obtained with the two variables being orthogonal. In the current illustration, U_{max}(x,y) is 3.09 bits. The difference between the maximum joint uncertainty, U_{max}(x,y), and the previous joint uncertainty, U(x,y), indicates the extent of correlation or redundancy existing between x and y. In Garner's nomenclature, this difference is called *contingent uncertainty*, U(x:y):

$$U(x:y) = U_{max}(x,y) - U(x,y) \quad (5)$$

In our example, U(x:y) is 0.28 bits. Because U(x,y) is always equal to or less than U_{max}(x,y), contingent uncertainty, U(x:y) is always non-negative. Again, due to its non-metric nature, it is best thought of as the amount of correlation between x and y. Unlike Pearson's correlation, for example, U(x:y) is not bounded by ± 1; it has no upper limit.

Using elementary, but fairly heavy-duty, algebra, Garner (1962) managed to extend this way of partitioning information to cases involving more than two variables. He worked out in full quantitative detail the mathematical structures sustaining the notions of *multivariate uncertainty* and *multidimensional information transmission*. In the development, a wealth of further new terms needed to be defined, including *partial contingent uncertainties, interaction uncertainties, higher order interactions, multivariate criteria, internal and external constraint, total constraint,* or *higher order total constraint*. This mathematical-conceptual edifice is completely general.

The last point is important. Garner (1962) provided a *formal analysis* of Shannon's (1948) information statistics. The analysis and results are "neutral," i.e., are amenable to various theoretical interpretations. Only when considered within a model of communication (or within other models or situations) do the terms assume theoretical meaning. Consider communication. Suppose that x in our first matrix is the source or input to a communication system, with known uncertainty distribution of the signals. The signals are transmitted through the system so that the receiver at the other end, y, records each signal upon arrival. This output, too, has its own distribution of uncertainties, but is it related to the input? Was the information in the input transmitted

Table 2 Matrix of expected probabilities of the joint occurrence of values of variables x and y from Table 1 assuming zero correlation between x and y

		y			
		1	2	3	
X	1	0.12	0.08	0.15	**0.35**
	2	0.08	0.06	0.10	**0.24**
	3	0.15	0.10	0.17	**0.41**
		0.35	**0.24**	**0.41**	

to the output? If so, to what extent? To answer this question, one needs to know the joint probability distribution of the input and output signals (i.e., the entries in the cells of the bivariate matrix) and then one must calculate the contingent uncertainty. Contingent uncertainty thus provides a measure of the *transmitted information*. Garner and Hake (1951; Hake & Garner, 1951) first suggested that humans can be considered as communication systems with x as the stimuli impinging on the human and y as the responses of the human to the stimuli. Consequently, contingent uncertainty has been widely used in psychology as an assay of transmitted information.

It is apparent that transmitted information cannot exceed the uncertainty of the input. For example, if the stimulus distribution has an uncertainty of 4 bits, transmitted information cannot be more than 4 bits. Consequently, transmitted information pertains only to a given experiment. To test whether there is a limit on transmitted information – channel capacity – one must perform a series of experiments with different input uncertainties. Very generally, it has been found that stimulus uncertainty larger than about 2.3 bits exceeds human's ability to discriminate the stimuli (Garner, 1962; Miller, 1956). As a result, beyond this value, information transmission remains constant regardless of stimulus uncertainty.

Notably, Garner (1962) pointed out that in other situations, contingent uncertainty is a measure of the extent to which information is *not* provided. This occurs most frequently with sequential behavior such as, for example, in language. Consider the letter q in a word. In principle, the next stimulus can be any one of the 26 letters of the alphabet, i.e., 4.7 bits of information. However, the letter q is always followed by the letter u, so that q conveys no information. We say that there is redundancy. In the words of Miller (1953, p. 6), "we encode about one-fourth as much information per letter as we might if we used our alphabet more efficiently. Our books seem to be about four times as long as necessary." The same situation occurs with whole words. Suppose that you need one word to complete the sentence: "Many experiments probing the nature of imagery were carried out over the last …" The semantic context of the sentence already constraints the number of alternatives from which the last word can come. The redundancy is considerable. In general, language entails very large amounts of redundancy at all lexical levels, which probably serves as a safety net for comprehending meaning.

Finally, we would be amiss if we did not mention an important byproduct of Garner's formalism: association of information and the mainstream statistical tool of analysis of variance (ANOVA). In a series of studies, Garner worked out in full technical detail the relation between ANOVA and information partitioning (Garner, 1953; Garner & Hake, 1951; Garner & McGill, 1956; McGill, 1954). There is a close affinity between the two given that both are based on the variance in the data and that both decompose the variance into separate sources. As a result, one can transform an information measure into the parallel ANOVA measure and vice versa. The main difference is that the predicted variable in ANOVA is metric, whereas the predicted variable in information analysis is nonmetric.

Sets, subsets, and redundancy

The information notion of redundancy plays a critical role in figure perception. The major clue is an often-overlooked result obtained within Garner's formalism: "selection of fewer stimuli than can be generated with a given number of variables produces a redundant set of stimuli" (Garner, 1962, p. 321). In other words, any subset drawn from a set of stimuli contains correlations between the values of the variables (subject to the proviso that the subset still contains the original variables). As we just recounted, correlation means redundancy or lack of information. The theorem is best understood by the visual illustration below (inspired by Garner, 1974).

The eight figures in the illustration (Fig. 3) are created by crossing the binary values of three variables: global form (circle, square), slope of the diagonal (upwards, downwards), and type of line (solid, broken). Because all possible combinations are realized in the illustration, the collection comprises a *total set* or population.

A ready selection of a sample would entail discarding all the circles, so that we are left only with squares. Note, however, that in this case the squares comprise another *total set* – which is now defined by two variables (slope of the diagonal and type of line). An example of a subset is given in the illustration below (Fig. 4).

The correlation between the variables is obvious in the subset. When the global form is of a solid line, the diagonal is of a positive slope, whereas when the global form is of a broken line, the diagonal is of a negative slope. This correlation or redundancy is not accidental: All subsets must contain correlation or redundancy. Moreover, the smaller the subset, the larger redundancy. In the extreme, the smallest subsets entail a very large amount of redundancy so that they convey no information.

The implications of this result are tremendous for perception – if only because each single figure (stimulus) is perceived along with the subset in which it is member. We elucidate this deduction in the next section but let us highlight here the main consequence: A stimulus that comes from a small subset carries the correlations and redundancy of its subset. Due to the redundancy, that stimulus conveys

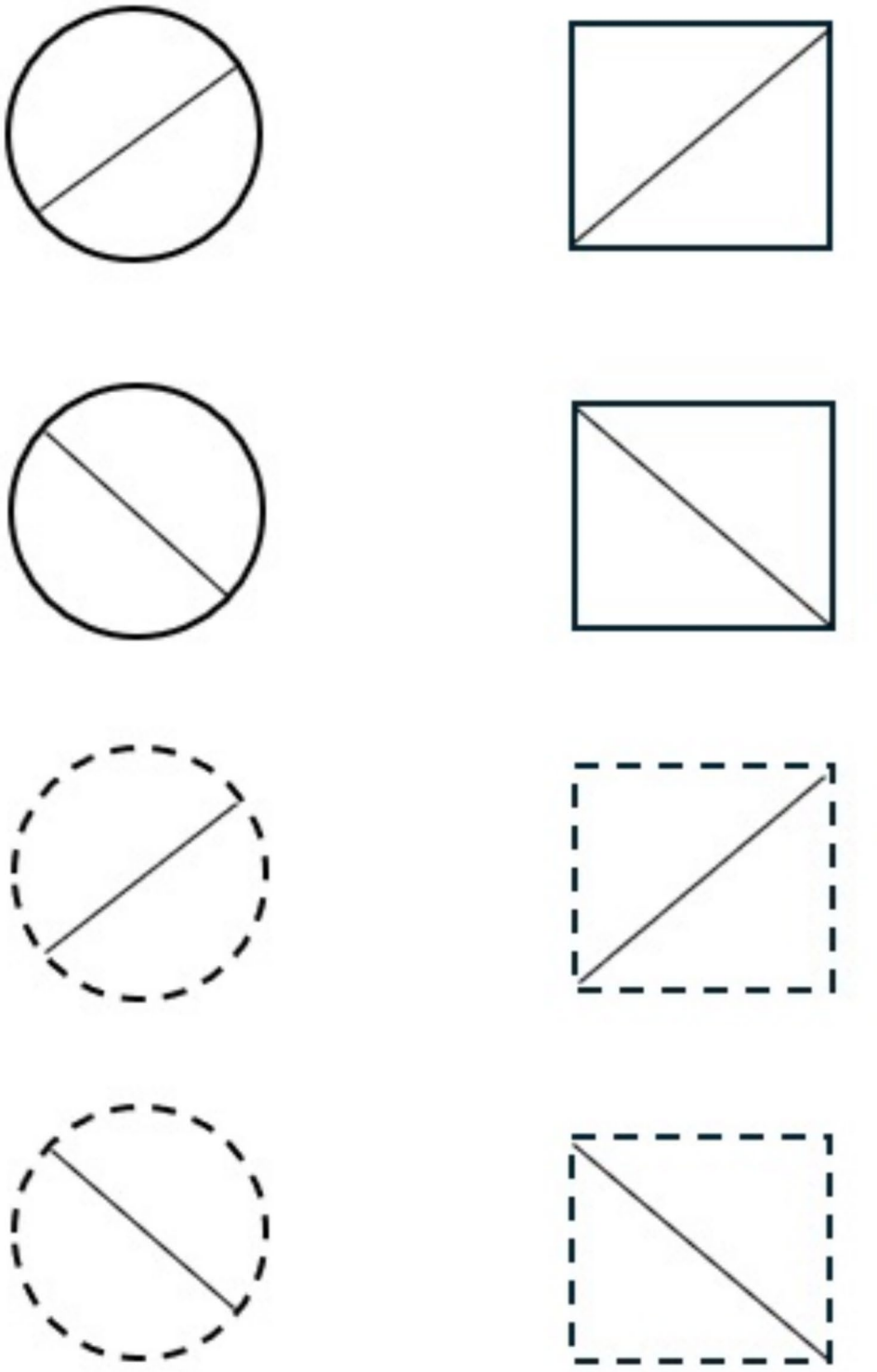


Fig. 3 A total set of eight stimuli formed by the binary values of three dimensions: form (circle, square) x slope (upwards, downwards) x line (solid, broken)



Fig. 4 A redundant subset of four stimuli. Notice the correlation between line and slope: with solid lines the slope is upward, whereas with broken lines the slope is downward. All subsets must contain redundancy or correlation

little information (see Fitousi & Algom, 2024, and Algom & Fitousi, 2025, for a detailed discussion of this idea).

Garner #4: Figural goodness, Gestalt, and the visual arts

Garner's main take from information theory was this: Perception of a stimulus depends on the alternative stimuli – those stimuli that could have appeared but did not in that instance. There is no such thing as perception of a single stimulus (apart from sensation of its physical energy). The stimulus derives its meaning from the subset of its alternatives – the N in the fundamental formula of information theory (see Eq. 1). A subset of alternatives always exists, otherwise a word (any stimulus) does not have meaning. In Garner's words, "The single stimulus has no meaning except in the context of alternatives…. the organism infers sets of stimulus alternatives, and without these inferred sets, no one can describe the single stimulus" (Garner, 1974, p. 185).

The challenge for the student of perception is recovering in each case the operational subset of alternatives. In everyday life, context often provides the relevant subset. When presented with the letter "e," the subset may well be the other vowels or perhaps all letters of the alphabet; with the numeral "2," the subset may include the neighboring numerals "1" and "3" or perhaps all single digits. The subset of alternatives shows what the stimulus is not (what it might have been). How about other stimuli, for example, simple geometric figures? Garner's idea was subjecting the figure to certain spatial transformations by way of identifying the inferred subset – rotating and reflecting the figure in steps of 90^0 around the vertical, horizontal, left, and right diagonal. Each new stimulus produced by this procedure is a member in the Rotation and Reflection (R & R) subset of alternatives. The first application of the R&R idea was to patterns of five dots placed in the imagery cells of a 3×3 matrix (Garner & Clement, 1963; see also, Lachmann, 2000, for an in-depth elucidation of the R&R idea). Two patterns with a different size of R & R subset are presented in Fig. 5.

Consider the pattern at the left of Fig. 5. Its R & R subset size is 1 – the pattern remains the same throughout all transformations. The pattern at the right, by contrast, has an R & R subset size of 4 – four different patterns emerge through the R & R transformations. Garner's development is similar to the notion of symmetry in mathematical group theory, to wit, "We say something is symmetrical, when it looks the same from more than one point of view" (Carter, 2009). In fact, symmetry group analysis in mathematics is isomorphic to Garner's R & R set theory; the respective sizes are inversely related to one another (Fitousi & Algom, 2024; Palmer, 1991). Garner's development focuses on sets of patterns, whereas group theory focuses on transformations of a

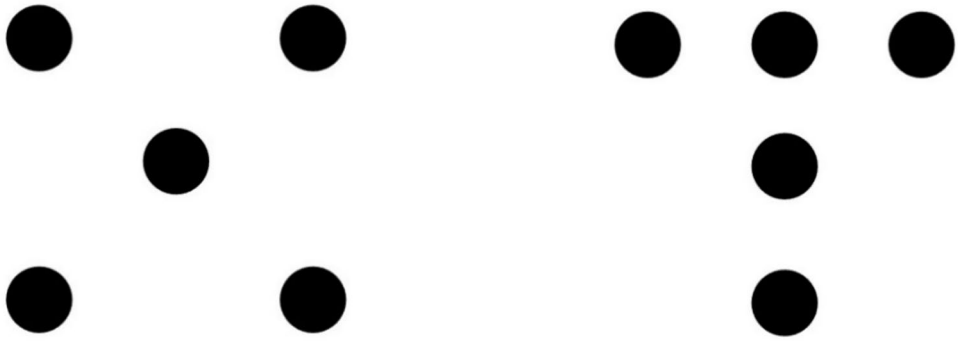


Fig. 5 Two patterns defined by their R & R subsets and by their symmetry groups. Left panel: Size of the R & R subset = 1; Symmetry group = 8 [I, V, H, L, R, 90°, 180°, and 270° rotations]. Right panel: Size of the R & R subset = 4; Symmetry group = 1 [I]. I: identity; V: vertical reflection; H: horizontal reflection; L: left-diagonal reflection; R: right-diagonal reflection

pattern. Considering the latter, one can say that the pattern at the left of Fig. 5 has more symmetries than the pattern at the right.

It is evident that the patterns of Fig. 5 also differ, as do all Gestalts, in their aesthetic appeal or in what the early Gestaltists called figural goodness. To tap attractiveness, the participants in the Garner and Clement (1963) study rated the patterns for figural goodness. In a separate task, the participants placed each pattern in a group with other patterns based on perceived similarity. The groups did not have to be of equal size. A sizeable correlation of 0.84 was found between figural goodness and similarity group size: Good figures tended to reside within small groups of similar patterns. And the very best figures were groupless – the size of their group of similar patterns was 1 (e.g., the left pattern of Fig. 5). This result is poignantly put in the title of Garner's (1970) article: "Good patterns have few alternatives," as well as in the article's conclusion (p. 42): "Poor patterns are those which are not redundant and thus have many alternatives, good patterns are those which are redundant and thus have few alternatives, and the very best patterns are those which are unique, having no perceptual alternatives."

Garner's results align well with the readily observable fact that virtually all common symbols in use by people at large are good figures. Public symbols are typically redundant, containing symmetry and further regularities. Most remain the same when viewed from different angles, i.e., when flipped around or rotated. Consequently, they do not saddle the perceptual system with extensive processing. They convey little information, hence they are perceived instantaneously and retained easily in memory (e.g., Roberts et al., 2023). Could that be the reason for including the left-hand pattern of Fig. 5 to represent five in the game of domino? Could the same factor (e.g., symmetry) explain the initial popularity of Hermann Rorschach's inkblot test? In Fig. 6 we reproduce several well-known images used by public institutions.

Among Garner's students, it has been James Pomerantz who has pursued the aesthetic, visual, and organization properties of simple Gestalts. Following the insights gained from an early study (Pomerantz & Garner, 1973), Pomerantz explicated the concept of configural processing and configural superiority, uncovering several stunning visual effects (e.g., Pomerantz, 1991, 2003, 2006). We return to discuss Pomerantz's ideas in the next and last section. More recently, we have shown that figural goodness and patterness (or their inverse, randomness or patternessless) are tightly associated with modern species of algorithmic complexity (Algom & Fitousi, 2025; Fitousi & Algom, 2024). We found that the larger the R & R subset of a dot pattern, the more complex it is in the outcome of algorithmic complexity programs. Subjectively good figures are associated with small amounts of complexity in the computational sense.

Garner #5: Independence in perception and selective attention

The notion of independence is as pivotal in psychology as it is in mathematics. In probability theory we say that events A and B are independent if the probability of occurrence of A given that B is also occurring is the same as of the probability of A given that B is not occurring (the same applies with respect to the probability of B given that A is or is not present). Hence, A and B are independent if:

$$P(A/B) = P(A), \tag{6}$$

which shows that the occurrence of B does not affect the probability of A (and vice versa). A little elementary algebra leads to the product rule definition of independence:



Fig. 6 Famous images of public institutions and systems. All include appreciable amounts of symmetry, hence convey little or no information. It is easy to perceive, recognize, and remember them

$$P(A \text{ and } B) = P(A) \times P(B), \tag{7}$$

meaning that A and B are independent if the probability of their joint occurrence equals the product of the separate probabilities of A and of B. We can transform this equation to one of information in bits (by computing log to the base of 2 with a negative sign for probability, which varies between 0 and 1) to get:

$$I(A \text{ and } B) = I(A) + I(B). \tag{8}$$

This means that the information of the joint event AB equals the simple linear sum of the information of the individual events A and B. A trivial corollary is that the amount of information of the joint event AB exceeds that of the single event A (or B) – if indeed A and B are independent. Hence for independent events,

$$I(A \text{ and } B) > I(A). \tag{9}$$

If, on the other hand, the events are not independent:

$$I(A \text{ and } B) = I(A), \tag{10}$$

so that the information of A is not affected by the added occurrence of B.

A formidable challenge for students of psychology has been to incorporate into their discipline the mathematical definitions of independence. Is it possible to realize in the cognitive domain the mathematical statements? If so, how? In yet another pioneering contribution, Garner and Morton (1969) tackled the issue and showed how to realize independence in experimental situations. In their instantiation, A may represent the dimension of interest so that the participant performs a task (e.g., identification) with respect to the values of A. B is a second dimension that can be added to the values of A (i.e., AB is presented) or can be removed from the values of A (only A is presented) – all the while the task is performed with respect to A. Parity in performance across AB and A would reveal that A and B are not independent, whereas better performance to AB than to A would reveal that A and B are independent in processing. According to a closely related criterion of parity, "two perceptual processes are considered independent if performance on two perceptual tasks carried out simultaneously is the same as the sum of the performances on each task carried out separately" (Garner & Morton, 1969, p. 235).

For an illustration, suppose that the participant is presented with eight lines of different length (A), one line at a time. The task is to identify each line by an appropriate label (e.g., the numbers 1–8). The lines are discriminably different from one another (i.e., all immediate neighbors are well above the difference threshold), so that the task is purely that of absolute identification. The typical participant can correctly identify about five of the lines, i.e., the transmitted information is between 2 and 3 bits (Miller, 1956). Suppose further that the experimenter subsequently adds color (B) to the stimuli, so that each line appears in a different color. Now the participant is able to correctly identify all eight lines (and could possibly identify additional lines if more lines were presented). This enhancement in transmitted information shows that A and B are independent in processing (to some extent). In other cases, joint presentation does not lead to improved performance. If one builds a rectangle

on each of the previous lines as the base, identification does not improve. This latter result, "the failure to find beneficial effects has usually elicited some surprise, because it has seemed so intuitively clear that redundancy is an asset for discrimination" (Garner & Lee, 1962, p. 367). However, precisely the partition between dimensions that do and that do not lead to improvement in performance, hence independence, proved vital in Garner's thinking and continues to inform cognition research (e.g., Algom, 2016; Algom et al., 2015).

These ideas were first evoked in Garner's, 1962 book (see also, Garner & Lee, 1962), but they were subsequently developed in full detail in a seminal work (Garner & Morton, 1969; see Fitousi, 2013, for further extension). Here we are able to provide only the gist of that analysis. The main conclusion, put in a grossly simplified way, was that parity in performance implies separable dimensions, whereas gain in performance implies integral dimensions. Garner used the terms *separable* and *integral* to indicate two modes of cognitive processing in lieu of the terms independent and dependent from probability theory. The integrality-separability distinction is a cornerstone of contemporary cognitive science (Algom & Fitousi, 2016).

Another major conclusion drawn from Garner and Morton's (1969) analysis was that independence is not a unitary concept. There are several experimental ways to approximate the notion, none of which is fully satisfactory. For example, zero correlation implies independence but only under special conditions; the same is true of dimensional orthogonality and even performance parity. Building on the foundations laid by Garner, the complex issue of independence was given a fresh look in the work of Ashby and Townsend (1986), aptly titled, "Varieties of perceptual independence." Rigorous definitions were given to all relevant terms including *perceptual independence*, *sampling independence*, and *decisional independence*, among others. A network of theorems connected all terms, resulting in a new formal *General Recognition Theory* (GRT). We cannot be technical except to note the inclusion of a judgment or decision component in GRT. The theory has been developed further to inform current research in cognition (e.g., Ashby, 1992; Ashby & Soto, 2015; Kadlec & Townsend, 2014; Soto et al., 2017).

These explorations of the notion of independence entailed tasks of identification, classification, card-sorting, and various forms of judgments. Subsequently, the introduction of speeded tasks led to a notable change in theoretical perspective and brought to the fore the notion of *selectivity of attention*. Given separable dimensions, each dimension is processed individually so that attention can be focused on one dimension without intrusions from the other dimension. In contrast, integral dimensions interact in processing so that it is impossible to selectively focus on one dimension and fully ignore the other. Selective attention is perfect with separable dimensions, but it is compromised with integral dimensions. Harnessing this logic, Garner (1974) created the benchmark design to tell separable and integral dimensions apart. This test, *Garner's speeded classification paradigm*, evolved into one of the most popular tools of cognitive science for assaying selective attention to any one dimension of the stimulus (Algom & Fitousi, 2016; see also Burns, 2014, 2016, for creative extensions of the basic paradigm).

The paradigm is depicted below with the dimensions of shape (A; circle, triangle) and color (D; red, green). Suppose that your task is to classify shape on each trial while ignoring the shape's color. In one block of trials, the task-irrelevant dimension of color is kept constant (all shapes are in red) so that only the target shape varies from trial to trial (Baseline). In another block of trials, the task-irrelevant color also varies in a random fashion (Filtering). Poorer shape performance in Filtering than in Baseline would show that selective attention to shape was compromised by variation of irrelevant color. In contrast, comparable performance would attest to perfect selectivity of attention to the target shape. The difference in performance between Baseline and Filtering defines the measure known as *Garner interference* (Pomerantz, 1983; see also, Algom & Fitousi, 2016). The complete Garner paradigm includes a third condition called Correlation. In Correlation, the irrelevant color also varies from trial to trial. However, unlike Filtering, it does so in tandem with target shape (e.g., the triangle appears in red on most or all trials). Because the irrelevant color is now predictive of the target shape, performance is often better than that in Baseline; this difference is called *redundancy gain* (Fig. 7).

As we just recounted, the standard interpretation holds that Garner interference signals the breakdown of selective attention to the target dimension. However, Pomerantz issued an important caveat that the difference between Baseline and Filtering sometimes records better performance in Baseline (see Fitousi, 2023; Niv et al., 2022). The stimuli in Baseline can sometimes generate emergent features that are conducive to superior performance. In these cases, the various Gestalts at Baseline, not the underlying dimensions determine performance. The original study of such Gestalts and the associated configural processing is that by Pomerantz and Garner (1973; see Fig. 8).

Pomerantz called such Gestalt-produced processing *configural superiority* and carried out painstaking research to study the effect. Rich rewards have been reaped. Across several decades, Pomerantz unearthed a stunning variety of compelling visual effects, illusions, and works of art (e.g., Pomerantz, 1991, 2003, 2006; Pomerantz & Portillo, 2011; Pomerantz & Pristach, 1989). Currently, configural processing and configural superiority figure prominently in the toolkit of students of cognition (e.g., Niv et al., 2022).

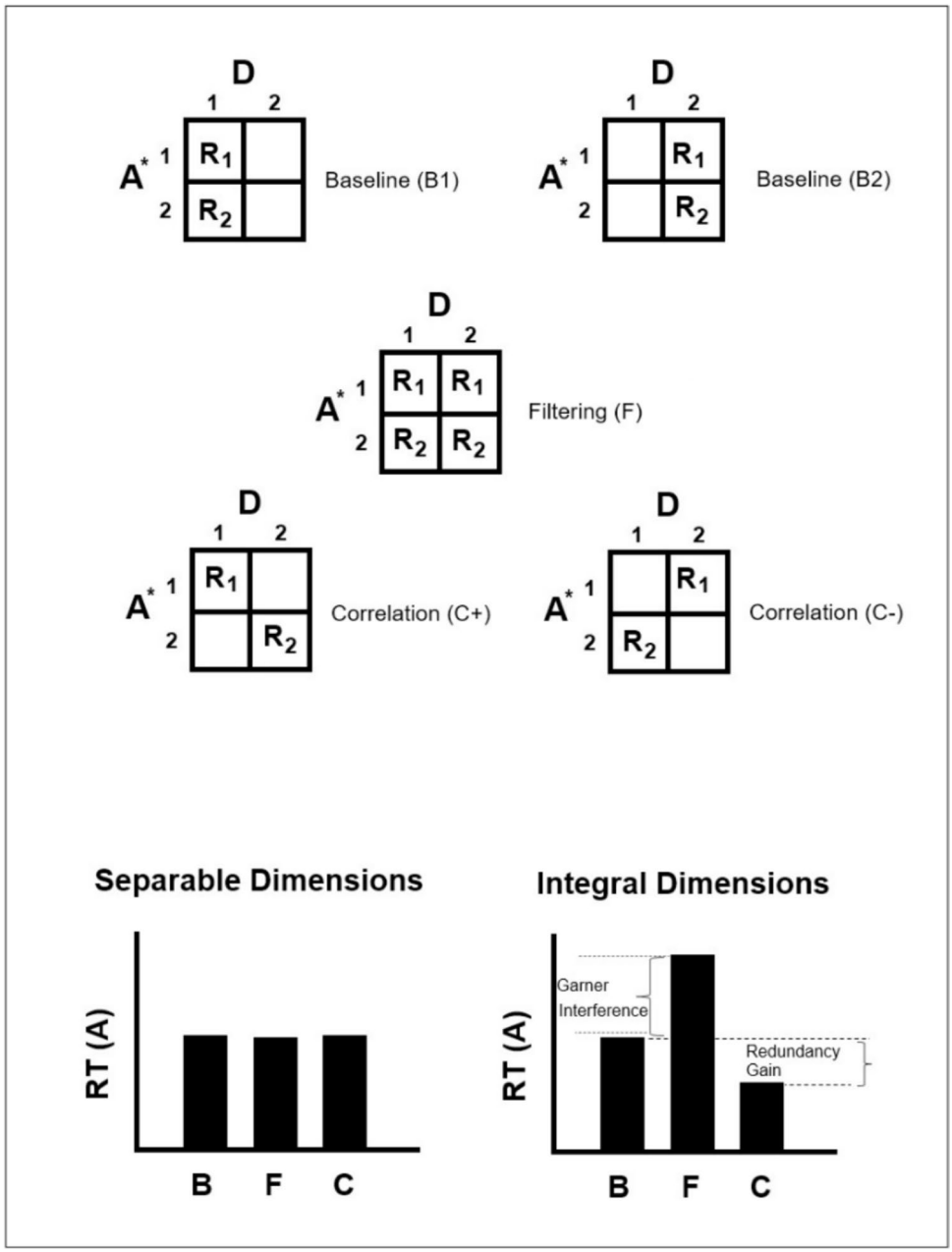


Fig. 7 Schematics of Garner's speeded classification paradigm. The three basic conditions are depicted on the top. The asterisk indicates that dimension A, not dimension D, was the relevant one for responding. R1 and R2 depict the correct responses to the two values of the relevant dimension A. The Correlation conditions depict two possible types of correspondence (positive, negative) in the appearance over trials of the values dimensions A and D. To connect to the example in the text, dimension A is shape and dimension D is color. Two prototypical outcomes, tapping integral and separable processing, are depicted at the bottom. B, F, and C, stand for Baseline, Filtering, and Correlation, respectively

If it is Pomerantz who carries the torch of the Garner paradigm, Gestalt, and configurality research, it is James Townsend who carries the torch of independence study through rigorous mathematical models. Consider the thought experiment suggested in the study of Garner and Morton (1969, p. 246): "the experimenters might carry out a threshold experiment with lights presented or not presented, in all possible combinations, to the two eyes separately but simultaneously. The subject is then required only to state whether he did or did not see a light… [also] stimuli varying in brightness could be used, but with the subject being required to give a single response" (parentheses added).

The real experiment was carried out some 30 years later by Townsend and Nozawa (1995). The experiment then

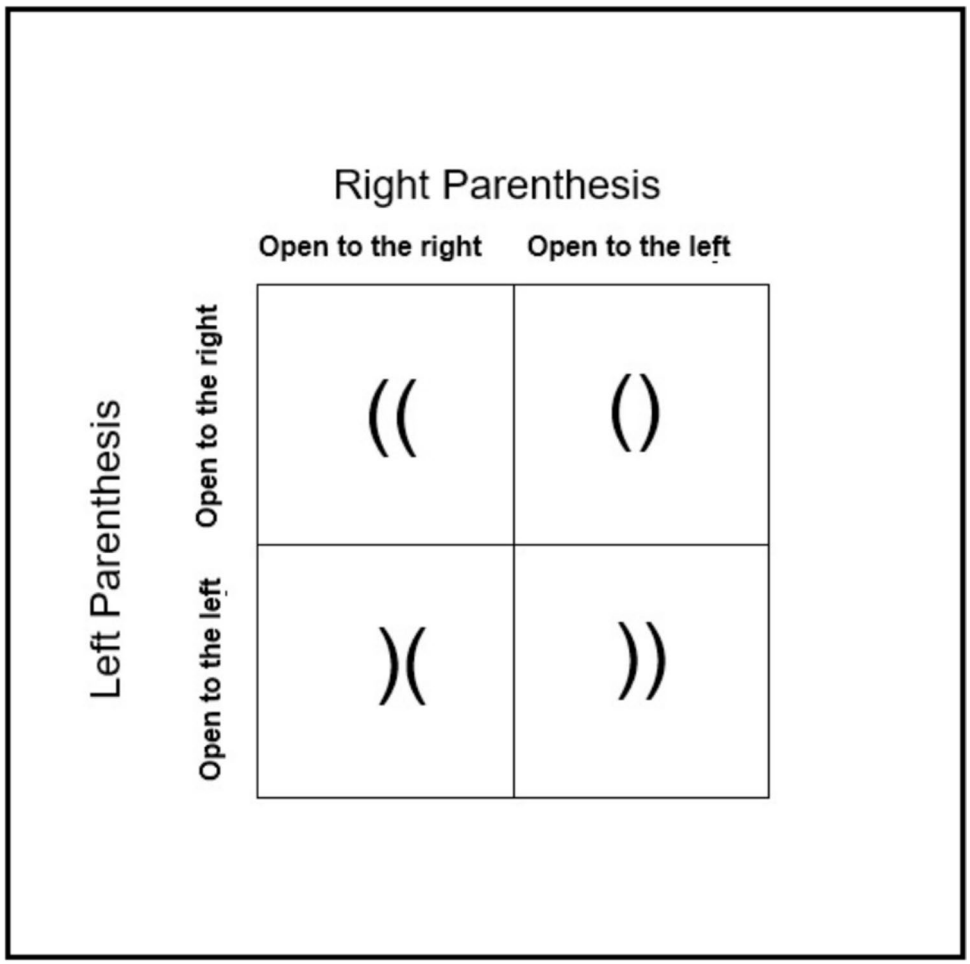


Fig. 8 Illustration of the stimuli presented in the classic study of Pomerantz and Garner (1973). The dimensions are (**A**) Left or Right parenthesis, each (**B**) with opening to the left or opening to the right. Notice the unique Gestalts with the emergent features of symmetry, parallelism or closure produced by the conjoined dimensional values in Baseline (e.g., the two stimuli in the top row)

served the basis for creating a network of definitions, theorems, and proofs that evolved into the immensely influential *Systems Factorial Technology* (SFT; e.g., Houpt et al., 2016, 2018; Little et al., 2017). Independence, the focal concept of Garner's explorations, is integrated as one component of a general theory of elementary perceptual processes. Properties of various information processing systems are identified not only by *(in)dependence* but also by *architecture*, *capacity*, and the operating *stopping rule* for emitting the response. The SFT proved a fertile ground for creative research in the gamut of current cognitive science from numerical cognition (Fitousi & Algom, 2018) to Stroop and Garner effects (Algom et al., 2017) to spatiotemporal perception (Lappin et al., 2016) to face (Fitousi & Wenger, 2013) and tone perception (Lentz et al., 2017) to decision making (Yang, 2017) – apart from advances in mathematical psychology per se.

The big picture

Offering a synopsis of Garner's contributions to psychological science is a challenging task not only due to the diverse contents but mainly because psychologists are still processing the contributions. Although Garner's ideas increasingly inform the course of cognitive research, his insights and contributions still have not been sufficiently digested by present-day investigators. Garner bequeathed a foundation to future development in several major fields, and our goal in this article was to further elucidate those foundations. This summary focuses on three major themes: loudness measurement, methodology and converging operations, and stimulus structure.

Garner's first domain of interest was the psychophysics of hearing. Young and irreverent, he was a trailblazer for several key concepts and techniques that are now standard in audition (e.g., scale development, cross-validation in *phon* units, parametrization of integration). Apart from his

Table 3 Major lines of research in cognition and perception building on the foundations by W. R. Garner

Foundations	Contemporary psychological science
Hearing	Auditory temporal integration
Psychophysics	Lambda scale of loudness
Methodology and Converging Operations	Resolution of the replicability and credibility crisis through multimethod validation
Psychological Information Theory	All modern applications of information theory
Figure Perception/Figural Goodness	Psychology of the arts, algebraic complexity
Independence in Perception	General Recognition Theory; System Factorial Technology
Selective Attention	Garner's speeded classification paradigm; Garner interference; configuration research

pioneering work on temporal integration in audition, perhaps the most important contribution concerned the measurement of loudness, and, by extension, of all psychological measurement. Garner insisted that any measurement must be sustained by a conceptual infrastructure – by a theory. This was a few decades before the publication of first volume of *Foundations of Measurement* by Krantz et al. in 1971, which made the theoretical requirement widely recognized. It is now roundly accepted that all measurements need conceptual support and that the validity of a measurement depends on the conceptual infrastructure supporting it. Psychophysicists were slow to recognize that the method of *magnitude estimation*, if it is to survive as a measurement, must be supported by a theoretical structure (Narens, 1996). Subsequently, Luce et al. (2010) developed a solid axiomatic structure for magnitude estimation (see also, Ellermeier et al., 2021). After seven decades, psychologists are still processing these ideas regarding proper psychological measurement.

Garner left an equally deep imprint on our methodology. Nowhere is this influence more striking than in the challenges posed by so-called replication crisis in psychology. The typical way of addressing the crisis has been imposing additional paperwork and bureaucratic burden on authors, mainly in the form of preregistration and other documentation. However, it is now recognized to an ever-greater extent that these rigors are conducive to a *theoretical crisis* in lieu of the replication crisis (Oberaurer & Lewandowsky, 2019). The reason is that preregistering a single method or a poor hypothesis does not advance broad rational reasoning and, indeed, stifles theoretical development. Garner's goal in advocating the converging operations idea was precisely that of avoiding a theoretical crisis. In Vallesi's (2025, p. 3) words, "theories and models need to consider a broad spectrum of variables and interaction," attainable through converging operations.

Arguably Garner's most impactful contribution concerns stimulus structure and information. The pertinent and quite radical ideas continue to shape cognitive science in meaningful ways. They are perhaps best captured by the title of Eleanor Rosch's review of Garner's, 1974 book, "The World is a Subset." The natural world surrounding the organism comprises a very large set of physical variables or dimensions. However, due to the limitation of the senses, the organism perceives only a subset of those dimensions. Now, according to the theorem that Garner proved and expounded, all subsets contain correlations among their constituent dimensions. It follows that our world contains correlation and redundancy – structure – whenever we perceive. The observer can use (or ignore) the structure that exists in the stimulus in different ways. Note, too, that structure and redundancy are properties of sets stimuli. What about single stimuli? Psychologically speaking there is not such an object as a single stimulus! A stimulus always implies its inferred subset, its alternatives. The properties of any stimulus are those of the (inferred) subset in which it exists. The smaller the subset, the more redundant and better (in the Gestalt sense) is the stimulus.

These ideas are supported by an evolved formalism coined in terms of information theory. For Garner, the basic teaching of information theory for psychology is that "information is a function not of what the stimulus is, but rather of what it might have been, of its alternatives" (Garner, 1974, p. 194).

Clearly, Wendell R. Garner's contribution to the science of psychology is enormous. The dramatic shift in his career, from psychophysics to cognition, can be viewed as part of a larger movement in psychology – away from behaviorism to cognition. Garner was a leader, along with colleagues, of this movement.

Coda

The research examined in this article is a few decades old. However, in this case past is not history. The methods and theories discussed are very much present in today's psychology. As is shown in Table 3, one can find precursors of many recent developments in Garner's ideas. The profound contributions of W.R. Garner are instrumental in shaping and molding contemporary research in perception and cognition.

Acknowledgements We thank Colin MacLeod, James Pomerantz, Bennett Schwartz, and an anonymous referee for very helpful comments on earlier versions. We are especially indebted to Colin Macleaod for his expert investment in refining the final product. The work was supported, in part, by an Israel Science Foundation grant to Daniel Fitousi (ISF-1498-21) and to Daniel Algom (ISF-543-19).

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